

ScamShield AI

Scammers isolate. ScamShield reconnects.

An accessibility-first AI cybersecurity command center that defends citizens against digital arrests, payment fraud, and high-pressure social engineering.

Hackathon Project Report

1. Project Overview

ScamShield AI is a web-based, accessibility-first safety platform built to protect users — especially senior citizens and first-time smartphone users — from digital scams such as fake "digital arrest" calls, UPI QR-code fraud, urgent KYC/SIM-blocking threats, and AI voice-cloning fraud. The application combines a deterministic rule-based scam detector with an optional local AI (Ollama) integration, an interactive training simulator, a trusted-contact "Safety Circle", and a community scam-reporting radar into a single React single-page application.

The project is implemented entirely as a client-side React + Vite application, with all user data (profiles, scan history, scores, contacts) stored on-device, and no dependency on any external cloud backend.

2. Problem Statement

Digital scammers rely on fear, manufactured urgency, impersonation of authority, and deliberate isolation of the victim to bypass rational judgement. Vulnerable groups — senior citizens, first-time digital users, and people unfamiliar with banking/UPI mechanics — are disproportionately targeted. The README documents four recurring patterns the project specifically targets:

- **Digital Arrest scams** — fraudsters impersonate police, CBI, or telecom regulators and threaten immediate arrest via fake video calls.
- **UPI QR fraud** — victims are tricked into entering a UPI PIN under the false belief that doing so will let them receive a refund.
- **Urgent KYC / SIM-blocking threats** — panic-inducing messages threaten to suspend banking or telecom services within hours.
- **AI voice-cloning fraud** — cloned voices simulate family emergencies to extract urgent money transfers.

Existing awareness efforts (posters, one-off advisories) are largely passive and do not intervene at the moment a scam attempt actually reaches a user, nor do they help the user reconnect with a trusted person before making a financial decision under pressure.

3. Proposed Solution

ScamShield AI's central idea is to build "Scam Immunity" — a measurable, on-device preparedness score — through a combination of real-time detection, guided practice, and social reconnection, rather than relying on the user remembering abstract safety tips under pressure.

- **Detect:** a Scam Scanner analyses pasted text, screenshots, or call-transcript content against a weighted set of scam-signal patterns (and, if the user runs a local Ollama model, an LLM-assisted read) to produce an explainable risk verdict.

- **Reconnect:** when a high-risk scam is detected, the app actively prompts the user to contact someone from their Safety Circle before acting — directly countering the isolation tactic that most scams depend on.
- **Train:** an interactive Simulation Lab lets users rehearse realistic scam scenarios (e.g. a fake CBI video-arrest call) in a safe, branching sandbox, with immediate explanations for each choice.
- **Measure:** a Scam Immunity Score, calculated from five weighted components, gives users and (in a real deployment) their families a concrete, evolving measure of resilience — decoupled from gamified XP so the score can't be inflated by simply clicking through content.

This turns scam education from a one-time, passive activity into an ongoing, measurable, and socially connected safety habit.

4. Key Features Implemented

All of the following are implemented in the codebase (React pages under `src/pages`, and logic under `src/services`):

Module	What it does
Scam Scanner	Accepts pasted text (SMS, call transcript, chat) and runs it through the <code>SIGNAL_RULES</code> regex engine plus an optional local Ollama LLM call, producing a risk score (0-100), a risk band (Low → Critical), and a plain-language explanation of every matched red flag.
Risk Scoring Engine	Combines weighted signal matches (authority impersonation, threats, urgency, OTP/UPI PIN requests, remote-access requests, guaranteed-returns pitches, suspicious links, credential requests) with overlap dampening so repeated similar signals don't over-inflate the score.
Scam Immunity Score	A composite 0-100 preparedness index built from five weighted components: Scam Awareness, Digital Behaviour, Payment Safety, Simulation Performance, and Response Readiness — kept separate from XP/streaks so the score reflects verified capability.
Guardian Mode	An accessibility toggle that filters the navigation to essential items, enlarges typography and buttons, and simplifies the interface for elderly or first-time smartphone users.
Safety Circle	Lets a user maintain a list of trusted contacts; when the scanner flags a high-risk scam, the app prompts the user to alert a trusted contact using an auto-generated template message, directly countering the isolation tactic scammers use.
Simulation Lab	Multi-stage, branching interactive scenarios (e.g. a simulated 'Digital Arrest' video call, a UPI refund scam) where each choice carries a score delta and an explanation, letting users practice recognising and resisting pressure tactics safely.
Learn / Missions	Short guided missions (e.g. 'Spot the fake CBI call', 'Secure your UPI app') that award component-specific score gains and unlock safety badges on completion.
AI Scam Coach	A conversational panel backed by a local knowledge base and optional Ollama integration that answers user questions about scam patterns and safe practices.
Report Scam / Extractor	A community-reporting form whose <code>reportExtractor.js</code> utility automatically pulls phone numbers, UPI IDs, and domains out of a pasted description and classifies the likely scam type.
Scam Radar	A dashboard visualizing anonymized report data as a regional heatmap of India and a set of trending scam categories, intended to reveal emerging or coordinated campaigns.
Onboarding Flow	A guided first-run flow that collects age range, language, digital-comfort level, and typical digital activities, and runs a short baseline assessment (<code>ONBOARDING_SCENARIOS</code>) to seed the initial Immunity Score.

Multilingual Guidance	Deterministic scam-response guidance (e.g. for CBI/digital-arrest, UPI PIN, screen-sharing, and investment scams) is available in both English and Hindi.
Privacy Page & On-device Storage	All user data — profile, score state, scan history, safety circle, Ollama configuration — is stored client-side via a storage abstraction (window.storage with a localStorage fallback), with no data leaving the device by default.

5. Technology Stack

Layer	Technology	Role
Frontend Framework	React 18 + Vite	Component-based SPA with fast dev/build tooling
Styling	Custom CSS (index.css)	Dark cyberpunk theme, glassmorphic cards, keyframe animations
Icons	lucide-react	Consistent iconography across dashboard and navigation
Charts / Visualization	Recharts + custom SVG	Immunity score trends, region heatmaps, campaign clustering
Local AI Engine	Ollama (llama3 / phi3) via aiProvider.js & ollama.js	Optional local LLM calls for scam-text analysis and the AI Coach
Deterministic Detection	Custom regex rule engine (scamEngine.js)	Signal matching for authority impersonation, OTP/UPI PIN requests, urgency, remote-access requests, etc.
Data Persistence	window.storage API with localStorage fallback (storage.js)	Stores onboarding profile, immunity score state, scan history, Ollama config on-device
Build Tooling	Vite, ESLint	Development server, production build, and code-quality checks

The project has no external backend server — the local Ollama integration is the only optional network dependency, and it runs entirely on the user's own machine via a Vite proxy (/api/ollama) with a direct localhost fallback.

6. System Working / Methodology

The application's workflow, as reflected in the project's own architecture diagram (README.md) and source code, follows this path:

- A first-run Onboarding Flow collects the user's age range, language, digital-comfort level, and typical digital activities, and runs a short baseline assessment to seed an initial Immunity Score.
- Suspicious text (SMS, chat, or call transcript) entered into the Scam Scanner is passed through the deterministic SIGNAL_RULES regex engine and, optionally, a local Ollama LLM call.
- Matched signals (e.g. authority impersonation, OTP/UPI PIN request, urgency, remote-access request) are combined with overlap dampening into a 0-100 risk score, mapped to a risk band (Low, Caution, Suspicious, High, Critical) with a plain-language explanation for each flag.
- If the risk is high, the user is prompted to alert a trusted contact from their Safety Circle, using an auto-filled message template.
- Separately, the Learn tab offers short missions and the Simulation Lab offers branching scam scenarios; completing these updates specific components of the Immunity Score (awareness, digital behaviour, payment safety, simulation performance, response readiness) and can unlock safety badges.

- Community-submitted incident reports are passed through a lightweight extractor (reportExtractor.js) that pulls out phone numbers, UPI IDs, and domains, classifies the likely scam type, and feeds the aggregated, anonymized data into the Scam Radar's regional heatmap and trend view.
- All state (profile, score breakdown, scan history, Ollama configuration) persists on-device through a storage abstraction that uses the window.storage API when available and falls back to the browser's localStorage.

7. Implementation Notes

- **Detection engine:** src/services/scamEngine.js defines the SIGNAL_RULES array (regex patterns with weights for signals such as authority impersonation, threats, isolation instructions, urgency, OTP/UPI PIN requests, remote-access requests, and suspicious links), the risk-band thresholds, and the runScamAnalysis() function that ties them together into a scored, explainable verdict.
- **AI integration:** src/services/aiProvider.js supplies deterministic, bilingual (English/Hindi) guidance for common scenarios (CBI/digital arrest, UPI PIN, screen-sharing, investment scams) and can hand off to src/services/ollama.js, which talks to a locally running Ollama instance (llama3 or phi3) through a Vite proxy, with a direct localhost fallback if the proxy is unavailable.
- **Scoring:** calculateImmunityScore() in scamEngine.js applies fixed weights (Awareness 25%, Digital Behaviour 20%, Payment Safety 20%, Simulation Performance 20%, Response Readiness 15%) to the five component values to produce the overall score, with helper functions for score deltas from simulations, missions, and scan rewards.
- **Report extraction:** src/services/reportExtractor.js uses regex-based parsing to pull phone numbers, UPI-style IDs, and domains out of free-text incident descriptions and infers a likely scam category from keywords.
- **Persistence:** src/services/storage.js wraps a window.storage key/value API (with a localStorage fallback) to persist onboarding data, score state, scan history, and Ollama configuration entirely on-device.
- **UI composition:** the React app (src/App.jsx) drives a single-page, state-based navigation bar (Dashboard, Scam Scanner, Simulation Lab, AI Scam Coach, Scam Radar, Safety Circle, Learn, Report Scam, Profile/Settings) that filters itself down to essential items when Guardian Mode is enabled, alongside reusable components (ScoreGauge, Badge, MissionModal, Toast, Card) and Recharts-based visualizations.

A built-in "3-Minute Hackathon Demo Flow" (documented in the README) walks a judge through loading a sample at-risk profile, scanning a scam SMS, enabling Guardian Mode, contacting a Safety Circle member, completing a Digital Arrest simulation, and viewing the Scam Radar — indicating the flow was deliberately designed for live demonstration.

8. Results / Expected Impact

As a functioning prototype, ScamShield AI demonstrates a complete, working loop from scam-text detection through score-based feedback to social reconnection and rehearsal — all running client-side with no external data dependency by default. The project does not include any external user-testing metrics or production usage numbers in the ZIP; the figures shown in the app (e.g. sample Immunity Score history, sample Safety Circle contacts, seeded regional heatmap counts) are illustrative demo/seed data used to make the interface demonstrable, not measured outcomes, and should be presented to judges as such.

Its practical usefulness lies in three groups it is designed for:

- **Elderly and first-time digital users**, via Guardian Mode's simplified, high-clarity interface and direct-dial safety shortcuts.
- **Families**, via the Safety Circle feature, which gives relatives a structured way to be looped in before a financial decision is made under pressure.
- **Community/awareness initiatives**, via the Scam Radar's aggregated view of emerging scam patterns by region.

9. Future Scope

The following are realistic extensions beyond what is currently implemented, and are not present in the current codebase:

- Real backend and account sync, so Safety Circle alerts and scan history can be shared across a user's own devices and with family members' devices, rather than remaining in single-browser local storage.
- Native mobile app (Android/iOS) with OS-level SMS/call screening integration, rather than requiring users to paste text manually into the Scam Scanner.
- Integration with an actual voice-cloning detection model to address the AI voice-cloning threat described in the README, which is currently covered only at the awareness/education level.
- Direct API integration with official reporting channels (e.g. cybercrime.gov.in / 1930) to submit reports automatically rather than only displaying the resource.
- Expanded language coverage beyond the current English/Hindi guidance, and audio read-aloud support across all pages for low-literacy users.

10. Conclusion

ScamShield AI addresses a pressing, real-world problem — social-engineering scams that exploit fear, urgency, and isolation — with a working, cohesive prototype rather than a single-feature demo. Its combination of an explainable, weighted risk-detection engine, an on-device Scam Immunity Score, an interactive simulation lab, and a Safety-Circle mechanism to actively counter victim isolation gives it a clear and differentiated value proposition for a hackathon setting. The client-side, privacy-preserving architecture and the built-in demo flow make it straightforward to present, while the identified future extensions (backend sync, mobile screening, voice-clone detection) give it credible room to grow beyond the hackathon.